Arpit Deosthale

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Professional Summary

AI-native Computer Engineering student with hands-on experience building and deploying machine learning systems, computer vision, and data-driven applications. Developed an ML-powered radar-based human detection system selected for the Smart India Hackathon Finals and built end-to-end AI solutions from data preprocessing to deployment.

Education

Dr. D.Y. Patil Institute of Technology, Pune 2023 – 2027
Bachelor of Computer Engineering

Technical Skills

Languages: Python, Java, C/C++, JavaScript
ML & Data Science: NumPy, Pandas, scikit-learn, TensorFlow/Keras, LangChain, Ollama, RAG Systems, Signal Processing, Anomaly Detection
Software Development: Flask, REST APIs, Node.js, Express.js, Android, Full-Stack Web Development
Databases: MongoDB, Firebase, PostgreSQL
Cloud & DevOps: AWS, Google Cloud Platform, Linux, Windows, Virtual Machines, Splunk SIEM
Developer Tools: Git/GitHub, Raspberry Pi, ESP32, Jupyter Notebook, Android Studio

Projects

VITAL Radar | ML-Based Human Detection System – SIH Finalist — Python, scikit-learn, Raspberry Pi, Flask, Firebase

- Owned the full product lifecycle end-to-end – problem framing, model training, real-time inference, and live demo – selected for Smart India Hackathon Finals from thousands of entries with no existing blueprint.
- Trained and deployed ML classification models on Raspberry Pi behind a Flask REST backend, achieving reliable real-time radar-based human detection under real-world conditions.
- Integrated Firebase real-time database and cloud storage for remote monitoring, delivering a fully functional, field-deployable prototype.

Wahan Mitra | Intelligent IoT Headlamp Automation — ESP32, IR/Ultrasonic/LDR Sensors, Embedded C

- Conceived, designed, and built a self-initiated IoT system on ESP32 with multi-sensor fusion (IR, ultrasonic, LDR) to solve a real low-visibility driving problem from scratch.
- Implemented real-time environment classification and automated beam-switching logic, applying embedded AI and sensor-driven inference with no reference template.

AutoAttend | AI-Powered Attendance System — Python, Flask, OpenCV, PostgreSQL, Firebase, Android

- Built a team-based, facial-recognition attendance system to replace manual, proxy-prone tracking, with a Java Android frontend and a Python/Flask backend.
- Designed REST APIs for real-time face detection, attendance marking, role-based access across student/teacher/admin roles, and Excel-based report exports.
- Used PostgreSQL and Firebase Realtime Database to manage authentication and attendance records across multiple user roles.

Legal Document Analyzer | LLM-Powered Document Intelligence — Python, LLMs, NLP, Retrieval-Augmented Generation

- Built an AI-powered platform that extracts, summarizes, and interprets legal documents using LLMs, reducing manual review effort on complex contracts and legal texts.
- Implemented document ingestion, semantic search, and context-aware question answering using NLP and retrieval techniques.

Experience & Leadership

Technical Event Organizer 2025
Accunetix 13.0 DSA Contest

- Co-organized a large-scale technical symposium, managing contest logistics, coordination, and problem-setting for the coding competition.

EPM Lead & Event Organizer 2025
TEDx

- Led the Event Production and Management (EPM) team, coordinating volunteers and ensuring seamless execution of the event.

Achievements & Certifications

Smart India Hackathon Finalist – Selected from thousands of teams with an independently built ML prototype (VITAL Radar)
TryHackMe – Top 5% Globally – 50+ hands-on labs in cybersecurity and AI-adjacent systems
20+ Hackathons & CTFs – Consistent competitor across domains; comfortable with ambiguity, pressure, and rapid pivoting
AWS & Google Cloud Certified (via Credly, ongoing) – Cross-domain cloud fluency supporting AI/ML deployments